

The Model with Private Signals

In the model of the draft, the signal on a firm is public: both banks see the same realization. This note solves the same model when each bank draws its own signal on the firm, with the same precision q . Given the firm's type the two signals are independent, so two banks can disagree about the same firm. The draft's public signal is the other extreme, in which the two signals always agree. The two models coincide when the signal carries no information ($q = \frac{1}{2}$) and when it is perfect ($q = 1$). Everything else follows the draft: search frictions, two types of firm, reservation rates, and the pooling and bad-only regions. The construction follows `note.pdf`; Section 8 lists where this note departs from it.

Main results.

1. After either signal, profit depends on two functions of the rate: what a bank earns on a good firm and on a bad firm. Their difference, the *premium* on a good firm, sorts the two signals. Banks that drew a good signal quote the lowest rates in the market, banks that drew a bad signal the highest (Section 3).
2. As in the draft, φ^s decides where bad-signal banks lend: in the pooling region only (full pooling), in both regions (hybrid), or in the bad-only region only. Within the pooling region, the rate ranges of the two signals meet, overlap, or are separated by a pause of the good-signal banks. Section 3.9 gives the algorithm, and Section 5 shows all four configurations in one economy.
3. A bank that uses less precision than its rival, or none, loses nothing whenever the two signals quote a common rate in equilibrium. With a cost of precision that rises with q , only two kinds of market sustain their precision: markets with no information, and markets in which bad-signal banks lend in the bad-only region only (Section 6).

1. Environment

Banks and firms. Two risk-neutral banks meet a continuum of firms. Each firm borrows one unit at rate r . A share π of firms are shoppers: they get an offer from both banks and accept the lower one. The other $1 - \pi$ are captive: they get one offer. A firm rejects any rate above its reservation rate.

Types. A firm is good ($\theta = g$, share μ_g) or bad ($\theta = b$, share $\mu_b = 1 - \mu_g$). Lending to it costs c_g or c_b ; costs do not depend on the rate, as in the calibration. Good firms accept rates up to $\bar{r}^g$ and bad firms up to $\bar{r}^b$, with

$$c_g < c_b < \bar{r}^g < \bar{r}^b.$$

Rates $r \leq \bar{r}^g$ form the **pooling region**, where both types accept. Rates $\bar{r}^g < r \leq \bar{r}^b$ form the **bad-only region**, where only bad firms accept.

Signals. Each bank $j \in \{1, 2\}$ draws its own signal $s_j \in \{g, b\}$ on the firm, with

$$\Pr(s_j = \theta \mid \theta) = q, \quad q \in (\tfrac{1}{2}, 1].$$

Given the firm's type, s_1 and s_2 are independent. Precision is the same for both types and for both banks.

One bank's signal. As in the draft, the shares of firms with each signal and the probability that the firm is good after each signal are

$$\sigma_g = q\mu_g + (1 - q)\mu_b, \quad \sigma_b = 1 - \sigma_g, \quad \alpha^{s_g} = \frac{q\mu_g}{\sigma_g}, \quad \alpha^{s_b} = \frac{(1 - q)\mu_g}{\sigma_b},$$

and the expected cost of lending after signal s is $c^{avg,s} = \alpha^s c_g + (1 - \alpha^s) c_b$. The signal sorts firms but does not change how many good firms a bank meets in total:

$$\sigma_g \alpha^{s_g} + \sigma_b \alpha^{s_b} = \mu_g. \tag{1}$$

Two banks' signals. Averaging over the type gives the joint distribution:

	rival drew g	rival drew b
bank drew g	$q^2\mu_g + (1 - q)^2\mu_b$	$q(1 - q)$
bank drew b	$q(1 - q)$	$(1 - q)^2\mu_g + q^2\mu_b$

The banks disagree on a share $2q(1 - q)$ of firms, whatever μ_g . For pricing, what matters is that given the firm's type the rival's signal does not depend on the bank's own: it is g with probability q for a good firm and $1 - q$ for a bad firm.

Example (used throughout): $\mu_g = 0.65$ and $q = 0.75$ give $\sigma_g = 0.575$, $\alpha^{s_g} = 0.848$ and $\alpha^{s_b} = 0.382$. The banks disagree on 37.5% of firms.

Strategies and equilibrium. A bank that drew signal s quotes from a rate distribution F^s with no atoms: F^{s_g} after a good signal, F^{s_b} after a bad one. A “good-signal bank” is a bank that drew a good signal on the firm at hand; each bank is a good-signal bank on some firms and a bad-signal bank on others. In a symmetric equilibrium both banks use the same pair, and the profit $\Pi^s(r)$ after signal s satisfies

$$\Pi^s(r) = K^s \text{ on the support of } F^s, \quad \Pi^s(r) \leq K^s \text{ at every } r.$$

Expected profit per firm examined is $\mathbb{E}[\Pi(q)] = \sigma_g K^{s_g} + \sigma_b K^{s_b}$.

Notation. A subscript g or b refers to the firm's type; a superscript s_g or s_b refers to the signal the bank drew.

symbol	meaning
$\alpha^s, c^{avg,s}$	probability that the firm is good, and expected cost, after signal s
F^s, K^s	rate distribution, and equilibrium profit, after signal s
$Y_\theta(r)$	probability that the rival's offer to a type- θ firm is above r
$\Phi_\theta(r)$	profit on a type- θ firm quoted r
$\Delta(r)$	premium on a good firm, $\Phi_g(r) - \Phi_b(r)$
$A^s(r), B^s(r)$	margin at stake on firms whose rival drew a good or a bad signal
φ^s	adverse selection after signal s , as in the draft
m	share of bad-signal quotes in the bad-only region
$\underline{r}, \underline{r}^b$	lowest rate in the market; lowest rate quoted in the bad-only region
$\rho, \rho_g, \rho_-, r_2$	rates that decide the configuration (Section 3.8)

2. Profit after each signal

2.1 The value of a good and of a bad firm

A shopper stays with a bank quoting r if the rival quotes more. The rival quotes from F^{sg} or F^{sb} depending on its own signal, and the distribution of that signal depends on the firm's type. The probability that the rival's offer to a type- θ firm is above r is

$$\begin{aligned} Y_g(r) &= q [1 - F^{sg}(r)] + (1 - q) [1 - F^{sb}(r)], \\ Y_b(r) &= (1 - q) [1 - F^{sg}(r)] + q [1 - F^{sb}(r)], \end{aligned}$$

and the bank's profit on a type- θ firm quoted r is

$$\Phi_\theta(r) = [1 - \pi + \pi Y_\theta(r)] (r - c_\theta). \quad (2)$$

Lemma 1 (profit). In the pooling region a bank that drew signal s earns

$$\Pi^s(r) = \alpha^s \Phi_g(r) + (1 - \alpha^s) \Phi_b(r), \quad r \leq \bar{r}^g.$$

In the bad-only region good firms reject, and $\Pi^s(r) = (1 - \alpha^s) \Phi_b(r)$.

Both signals face the same two functions and differ only in the weights. The lemma uses the independence of the signals given the type: once the type is known, the bank's own signal says nothing more about the rival's.

The same profit, counted by the rival's signal. A bank wins shoppers mostly when the rival drew the other signal, and those firms are worse than the bank's own signal suggests. The chance of winning and the margin therefore move together across types, so profit cannot be computed as an average chance of winning times an average margin. Take the 575 firms with a good signal out of 1,000 in the example, at $r = 0.40$, with $\pi = 0.4$, $c_g = 0.05$ and $c_b = 0.30$, when every good-signal quote lies below r and every bad-signal quote above. The bank keeps a firm with probability 0.6 if the rival drew g and with probability 1 if it drew b . The margin is 0.35 on a good firm and 0.10 on a bad one. Each cell is (firms) $\times$ (chance of keeping) $\times$ (margin):

	rival drew g	rival drew b	total
good firm	$365.6 \times 0.6 \times 0.35 = 76.8$	$121.9 \times 1 \times 0.35 = 42.7$	119.4
bad firm	$21.9 \times 0.6 \times 0.10 = 1.3$	$65.6 \times 1 \times 0.10 = 6.6$	7.9
total	78.1	49.2	127.3

- *Adding by rows* is Lemma 1. Within a row the margin is fixed and only the chance of keeping is averaged: 0.7 for good firms and 0.9 for bad firms, which are $1 - \pi + \pi Y_g$ and $1 - \pi + \pi Y_b$. Profit per firm is $0.848 \times 0.7 \times 0.35 + 0.152 \times 0.9 \times 0.10 = 0.221$.
- *Adding by columns* weights the rival's signal. The rival agrees with probability $387.5/575 = 0.674$, and then the average margin is 0.336. It disagrees with probability 0.326, and then the average margin is 0.2625. Profit per firm is $0.674 \times 0.6 \times 0.336 + 0.326 \times 1 \times 0.2625 = 0.221$.

2.2 Compact form

Substituting Y_g and Y_b into Lemma 1 gives, in the pooling region,

$$\Pi^s(r) = (r - c^{avg,s}) - \pi [A^s(r) F^{sg}(r) + B^s(r) F^{sb}(r)], \quad (3)$$

$$\begin{aligned} A^s(r) &= \alpha^s q (r - c_g) + (1 - \alpha^s)(1 - q) (r - c_b), \\ B^s(r) &= \alpha^s (1 - q) (r - c_g) + (1 - \alpha^s) q (r - c_b). \end{aligned}$$

The bank starts from the profit it would earn if every firm accepted, $r - c^{avg,s}$. Good-signal quotes placed below r take away shoppers whose rival drew g ; A^s is the margin at stake on those firms, and B^s is the same for rivals that drew b . Profit is linear in the two distributions. Two numbers recur below:

$$L = 1 - \pi q, \quad M = 1 - \pi + \pi q,$$

the chances of keeping a good firm and a bad firm when every good-signal quote lies below r and every bad-signal quote above.

2.3 The two limits in the draft

- **No information**, $q = \frac{1}{2}$. Both signals carry the belief μ_g and $Y_g = Y_b$. Lemma 1 becomes the draft's profit with no information: $[1 - \pi + \pi(1 - F(r))](r - c^{avg})$ in the pooling region and $\mu_b[1 - \pi + \pi(1 - F(r))](r - c_b)$ in the bad-only region.
- **Perfect signal**, $q = 1$. $\alpha^{sg} = 1$, $\alpha^{sb} = 0$, $Y_g = 1 - F^{sg}$ and $Y_b = 1 - F^{sb}$. Each signal competes only with the same signal, and the model splits into the draft's two one-type markets.

Section 4 shows that the equilibria converge to the draft's as well.

3. Equilibrium

Condition (A). Good-signal banks do not prefer to lend to bad firms only:

$$K^{sg} \geq (1 - \alpha^{sg})(1 - \pi)(\bar{r}^b - c_b).$$

A sufficient condition, which can be checked before solving, is $\varphi^{sg} \geq 0$, with φ^s defined in (6). Everything below assumes (A). When (A) fails, banks with either signal lend in the bad-only region; that case is not solved here (Section 4 gives an example). Proof sketches are in Appendix A.

3.1 The premium sorts the two signals

Subtracting the two profits in Lemma 1,

$$\Pi^{sg}(r) - \Pi^{sb}(r) = (\alpha^{sg} - \alpha^{sb}) \Delta(r), \quad \Delta(r) \equiv \Phi_g(r) - \Phi_b(r). \quad (4)$$

Δ is the **premium on a good firm**, and it is the only thing on which the two signals disagree. A good-signal bank prefers rates at which the premium is high, a bad-signal bank rates at which it is low. Write $\Delta^* \equiv (K^{sg} - K^{sb})/(\alpha^{sg} - \alpha^{sb})$.

Lemma 2 (sorting). Where F^{sg} has mass, $\Delta \geq \Delta^*$. Where F^{sb} has mass, $\Delta \leq \Delta^*$. Where both have mass, $\Delta = \Delta^*$, and Φ_g and Φ_b are each constant:

$$\Phi_g = \frac{(1 - \alpha^{sb})K^{sg} - (1 - \alpha^{sg})K^{sb}}{\alpha^{sg} - \alpha^{sb}}, \quad \Phi_b = \frac{\alpha^{sg}K^{sb} - \alpha^{sb}K^{sg}}{\alpha^{sg} - \alpha^{sb}}.$$

The last clause does the work: the two signals can quote the same range of rates only if neither the value of a good firm nor the value of a bad firm changes across it.

3.2 How the premium moves

Lemma 3. With $G_1(r) = q(r - c_g) - (1 - q)(r - c_b)$ and $G_2(r) = (1 - q)(r - c_g) - q(r - c_b)$,

$$\Delta(r) = (c_b - c_g) - \pi [F^{sg}(r) G_1(r) + F^{sb}(r) G_2(r)].$$

$G_1 > 0$, so good-signal quotes placed below r always lower the premium at r . G_2 is positive below $r^\dagger = [qc_b - (1 - q)c_g]/(2q - 1)$ and negative above it, so bad-signal quotes lower the premium at low rates and raise it at high rates. At the lowest rate in the market nobody has quoted yet, and the premium takes its largest value, $c_b - c_g$.

3.3 No rate in the pooling region goes unquoted

Lemma 4. In the pooling region, the union of the two supports has no gap.

If nobody quoted on an interval (a, b) with quotes on both sides, a bank quoting a could move to just below b : it would face the same rival quotes and charge more.

3.4 Who quotes the lowest and the highest rates

Proposition 1. $\Delta^* < c_b - c_g$, and only good-signal banks quote the lowest rate in the market.

Proposition 2. The highest rate quoted after a good signal is no higher than the highest rate quoted after a bad signal.

3.5 The top of the market: φ^{s_b}

At the highest rate in the market no rival quotes higher, so the bank earns only on captive firms.

Proposition 3. The highest rate in the market is $\bar{r}^g$ or $\bar{r}^b$, and

$$K^{s_b} = \begin{cases} (1 - \pi)(\bar{r}^g - c^{avg, s_b}) & \text{if } \varphi^{s_b} \geq 0, \\ (1 - \alpha^{s_b})(1 - \pi)(\bar{r}^b - c_b) & \text{if } \varphi^{s_b} < 0, \end{cases} \quad (5)$$

where, as in the draft,

$$\varphi^s \equiv (\bar{r}^g - c^{avg, s}) - (1 - \alpha^s)(\bar{r}^b - c_b) = \alpha^s(\bar{r}^g - c_g) - (1 - \alpha^s)(\bar{r}^b - \bar{r}^g). \quad (6)$$

In the second form c_b drops out: a bad firm is served in both regions, so its cost cancels. Quoting $\bar{r}^g$ keeps the good firms, with a margin $\bar{r}^g - c_g$ on each. Quoting $\bar{r}^b$ gives them up for the extra margin $\bar{r}^b - \bar{r}^g$ on bad firms. So $\varphi^s \geq 0$ exactly when the odds that the firm is good after signal s , $\alpha^s/(1 - \alpha^s)$, are at least $(\bar{r}^b - \bar{r}^g)/(\bar{r}^g - c_g)$.

3.6 The bad-only region

Proposition 4. Under (A), only bad-signal banks quote in the bad-only region. The share of their quotes there is

$$m = \min \left\{ 1, \max \left\{ 0, \frac{-(1 - \pi) \varphi^{s_b}}{\pi B^{s_b}(\bar{r}^g)} \right\} \right\}. \quad (7)$$

They quote on $[\underline{r}^b, \bar{r}^b]$, with

$$F^{s_b}(r) = 1 - \frac{1 - \pi}{\pi q} \cdot \frac{\bar{r}^b - r}{r - c_b}, \quad \underline{r}^b = \frac{(1 - \pi) \bar{r}^b + \pi q m c_b}{1 - \pi + \pi q m},$$

and nobody quotes on $(\bar{r}^g, \underline{r}^b)$.

In the bad-only region every good-signal quote lies below, so a bad-signal bank keeps a shopper only if the rival also drew b (probability q for a bad firm) and quotes higher. Profit falls at $\bar{r}^g$, where the good firms leave, and climbs back to K^{s_b} only at $\underline{r}^b$.

The draft's three regimes, after a bad signal. Equation (7) sorts bad-signal banks as the draft sorts a market with no information:

regime	condition	bad-signal banks lend in
full pooling, $m = 0$	$\varphi^{s_b} \geq 0$	the pooling region only
hybrid, $0 < m < 1$	$-\frac{\pi}{1-\pi} B^{s_b}(\bar{r}^g) < \varphi^{s_b} < 0$	both regions
bad-only, $m = 1$	$\varphi^{s_b} \leq -\frac{\pi}{1-\pi} B^{s_b}(\bar{r}^g)$	the bad-only region only

The threshold uses $B^{s_b}(\bar{r}^g)$, the margin on firms whose rival also drew a bad signal. If the rival always drew the same signal, as with the draft's public signal, $B^{s_b}(\bar{r}^g)$ would be the draft's $\bar{r}^g - c^{avg, s_b}$.

3.7 Rates both signals quote

By Lemma 2, Φ_g and Φ_b are constant where both signals quote. Their values are set at $\bar{r}^g$, where every good-signal quote lies below and a share m of bad-signal quotes lies above:

$$\Phi_g^\top = [1 - \pi + \pi(1 - q)m] (\bar{r}^g - c_g), \quad \Phi_b^\top = [1 - \pi + \pi qm] (\bar{r}^g - c_b).$$

Proposition 5. Where both signals quote,

$$\begin{aligned} F^{s_g}(r) &= \frac{1}{\pi} \left[1 - \frac{1}{2q-1} \left(\frac{q\Phi_g^\top}{r-c_g} - \frac{(1-q)\Phi_b^\top}{r-c_b} \right) \right], \\ F^{s_b}(r) &= \frac{1}{\pi} \left[1 - \frac{1}{2q-1} \left(\frac{q\Phi_b^\top}{r-c_b} - \frac{(1-q)\Phi_g^\top}{r-c_g} \right) \right], \end{aligned} \tag{8}$$

and $K^s = \alpha^s \Phi_g^\top + (1 - \alpha^s) \Phi_b^\top$. Where good-signal banks quote alone, below every bad-signal quote,

$$F^{s_g}(r) = \frac{(r - c^{avg, s_g}) - K^{s_g}}{\pi A^{s_g}(r)}, \quad \underline{r} = c^{avg, s_g} + K^{s_g}. \tag{9}$$

When $m = 0$, each type of firm faces, on a shared range, the rate distribution of a one-type market with cost c_θ and top rate $\bar{r}^g$: $Y_\theta(r) = \frac{1-\pi}{\pi} \cdot \frac{\bar{r}^g - r}{r - c_\theta}$. The noise in the signals does not change the rates a type faces there; it changes which signal supplies them.

Proposition 6. A bank cannot withdraw quotes as the rate rises, so on a shared range both distributions must be non-decreasing. On (8), F^{s_b} always rises. F^{s_g} falls below

$$\rho_- = \frac{c_b - \tau c_g}{1 - \tau}, \quad \tau = \sqrt{\frac{(1-q)\Phi_b^\top}{q\Phi_g^\top}},$$

rises above it, and equals one at $\bar{r}^g$. A shared range therefore ends at $\bar{r}^g$ and starts no lower than ρ_- .

3.8 Four configurations

Let $K^\circ = \alpha^{s_g} \Phi_g^\top + (1 - \alpha^{s_g}) \Phi_b^\top$, the profit of a good-signal bank at $\bar{r}^g$.

(iv) **Bad-only, if $m = 1$.** Bad-signal banks quote only in the bad-only region. Good-signal banks quote alone on the whole range $[r, \bar{r}^g]$, from (9) with $K^{sg} = K^\circ$, and reach one exactly at $\bar{r}^g$.

If $m < 1$, two rates decide the configuration.

- ρ is where bad-signal banks would start quoting on the shared solution: $F^{sb}(\rho) = 0$ in (8). It is the root in $(c_b, \bar{r}^g)$ of

$$q [(r - c_b) - \Phi_b^\top] (r - c_g) = (1 - q) [(r - c_g) - \Phi_g^\top] (r - c_b).$$

- ρ_g is where good-signal banks quoting alone at profit K° would have placed all their quotes: $F^{sg}(\rho_g) = 1$ in (9), that is

$$\rho_g = \frac{K^\circ + \alpha^{sg} L c_g + (1 - \alpha^{sg}) M c_b}{\alpha^{sg} L + (1 - \alpha^{sg}) M}.$$

Proposition 7. With $m < 1$, the pooling region takes one of the following three configurations, and no other.

(i) **Ranges meet, if $\rho_g < \rho$.** Good-signal banks would finish before bad-signal banks start, leaving rates nobody quotes, which Lemma 4 rules out. Instead K^{sg} rises above K° until good-signal quotes end exactly where bad-signal quotes begin. At that rate $F^{sg} = 1$ and $F^{sb} = 0$, so

$$\rho = \frac{K^{sb} + \alpha^{sb} L c_g + (1 - \alpha^{sb}) M c_b}{\alpha^{sb} L + (1 - \alpha^{sb}) M}, \quad K^{sg} = \alpha^{sg} L (\rho - c_g) + (1 - \alpha^{sg}) M (\rho - c_b).$$

Good-signal banks quote on $[r, \rho]$ from (9). Bad-signal banks quote on $[\rho, \bar{r}^g]$ with

$$F^{sb}(r) = \frac{\alpha^{sb} L (r - c_g) + (1 - \alpha^{sb}) M (r - c_b) - K^{sb}}{\pi B^{sb}(r)},$$

which reaches $1 - m$ at $\bar{r}^g$. The difference $K^{sg} - K^\circ > 0$ is a rent: good-signal banks hold a band of rates that bad-signal banks do not contest.

(ii) **Ranges overlap, if $\rho_g \geq \rho$ and $\rho \geq \rho_-$.** $K^{sg} = K^\circ$. Good-signal banks quote alone on $[r, \rho]$ from (9), and both signals quote on $[\rho, \bar{r}^g]$ from (8).

(iii) **Good-signal banks pause, if $\rho_g \geq \rho$ and $\rho < \rho_-$.** $K^{sg} = K^\circ$. Sharing from ρ would require good-signal banks to withdraw quotes. Instead F^{sg} stays at $\bar{F} = F^{sg}(\rho)$ until (8) climbs back to that level, at $r_2 \in (\rho_-, \bar{r}^g)$. On the pause $[\rho, r_2]$ only bad-signal banks quote,

$$F^{sb}(r) = \frac{(r - c^{avg, sb}) - K^{sb} - \pi \bar{F} A^{sb}(r)}{\pi B^{sb}(r)},$$

and both signals quote on $[r_2, \bar{r}^g]$ from (8). The pause is a band of rates never quoted after a good signal, although lower and higher rates are.

3.9 Algorithm

1. From (μ_g, q) : σ_g , σ_b , α^s and $c^{avg, s}$ (Section 1).

2. φ^{s_b} from (6), then K^{s_b} from (5).
3. m , $\underline{r}^b$ and F^{s_b} in the bad-only region, from (7).
4. $\Phi_g^\top$, $\Phi_b^\top$ and K° . If $m = 1$: configuration (iv).
5. ρ and ρ_g . If $\rho_g < \rho$: configuration (i); recompute ρ and K^{s_g} .
6. Otherwise ρ_- : configuration (ii) if $\rho \geq \rho_-$, configuration (iii) if not (then r_2).
7. Check (A).

The function `solve` in `simulate_equilibrium.py` follows these steps.

3.10 Worked example

$\mu_g = 0.65$, $q = 0.75$, $c_g = 0.05$, $c_b = 0.30$, $\pi = 0.4$, $\bar{r}^g = 0.45$, $\bar{r}^b = 0.80$.

step	quantity	value
1	$\sigma_g; \alpha^{s_g}, \alpha^{s_b}; c^{avg, s_g}, c^{avg, s_b}$	0.575; 0.848, 0.382; 0.088, 0.204
2	$\varphi^{s_b}; K^{s_b} = (1 - \alpha^{s_b})(1 - \pi)(\bar{r}^b - c_b)$	-0.063; 0.185
3	$B^{s_b}(\bar{r}^g); m; \underline{r}^b$	0.108; 0.881 (hybrid); 0.647
4	$\Phi_g^\top, \Phi_b^\top; K^\circ$	0.275, 0.130; 0.2531
5	$\rho; \rho_g$ (with $L = 0.7$, $M = 0.9$)	0.4441; 0.4433 < ρ : configuration (i)
5	meeting rate $\rho; K^{s_g}$	0.4437; 0.2534
7	(A): $K^{s_g} \geq (1 - \alpha^{s_g})(1 - \pi)(\bar{r}^b - c_b) = 0.046$	holds; $\varphi^{s_g} = 0.286$

In this equilibrium:

- Good-signal banks quote on $[0.341, 0.444]$.
- Bad-signal banks quote on $[0.444, 0.45]$, where F^{s_b} reaches $1 - m = 0.119$, and on $[0.647, 0.80]$. Nobody quotes on $(0.45, 0.647)$.
- Expected profit per firm examined is $\mathbb{E}[\Pi(q)] = 0.224$.

4. The limits: the draft's two benchmarks

No information. At $q = \frac{1}{2}$ both signals carry the belief μ_g , one distribution serves both, and Lemma 1 gives the draft's profit with no information, so the equilibrium is the draft's. As q falls to $\frac{1}{2}$ the equilibrium of Section 3 converges to it: in the economy of Section 5 (full pooling at $q = \frac{1}{2}$), $K^{s_g} = 0.18752$ and $K^{s_b} = 0.18749$ at $q = 0.5001$, against the draft's $(1 - \pi)(\bar{r}^g - c^{avg}) = 0.1875$. Near $q = \frac{1}{2}$, condition (A) depends on the draft's regime at $q = \frac{1}{2}$. At $q = 0.52$ it holds in every random economy checked with full pooling, in most with hybrid lending, and in none with bad-only lending (Section 7). In those economies the equilibrium is not solved here.

Perfect signal. As $q \rightarrow 1$, $\alpha^{s_g} \rightarrow 1$, $\alpha^{s_b} \rightarrow 0$, and $Y_\theta \rightarrow 1 - F^{s_\theta}$: each signal competes only with itself, as in the draft's perfect-signal equilibrium. Good-signal banks earn $K^{s_g} \rightarrow (1 - \pi)(\bar{r}^g - c_g)$ and

quote the draft's good-type distribution. After a bad signal, $\varphi^{sb} \rightarrow \bar{r}^g - \bar{r}^b < 0$ and $B^{sb}(\bar{r}^g) \rightarrow \bar{r}^g - c_b$, so

$$m \rightarrow \min \left\{ 1, \frac{(1 - \pi)(\bar{r}^b - \bar{r}^g)}{\pi(\bar{r}^g - c_b)} \right\}, \quad \underline{r}^b \rightarrow \max \{ \bar{r}^g, c_b + (1 - \pi)(\bar{r}^b - c_b) \}.$$

Bad-signal banks quote the draft's bad-type distribution on $[c_b + (1 - \pi)(\bar{r}^b - c_b), \bar{r}^b]$. When the lowest of these rates lies below $\bar{r}^g$, $m < 1$ in the limit: with no good firms left in their pool, bad-signal banks lose nothing at $\bar{r}^g$, and the band nobody quotes closes. So $m = 1$ in the limit only if $(1 - \pi)(\bar{r}^b - \bar{r}^g) \geq \pi(\bar{r}^g - c_b)$.

5. Simulation

The script `simulate_equilibrium.py` (same folder) computes the equilibrium with the algorithm of Section 3.9. At every parameter value it reports, it checks the equilibrium by brute force: on a fine grid of rates, profit from Lemma 1 equals K^s on each support and is no higher anywhere else. It writes the figures below to `figures/` and the results to `simulation_results.csv`.

Economy. $\mu_g = 0.65$, $c_g = 0.05$, $c_b = 0.30$, $\pi = 0.4$, $\bar{r}^g = 0.45$, $\bar{r}^b = 0.60$. Only the precision q varies. Condition (A) holds at every q in $[0.51, 0.99]$, and the brute-force check never finds an error above 10^{-15} .

5.1 Four equilibria

q	configuration	φ^{sb}	m	K^{sg}	K^{sb}	$\mathbb{E}[\Pi(q)]$
0.70	(i) ranges meet	0.094	0	0.217	0.157	0.191
0.80	(iii) good-signal banks pause	0.024	0	0.222	0.138	0.188
0.90	(ii) ranges overlap	-0.056	0.706	0.244	0.149	0.208
0.97	(iv) bad-only	-0.120	1	0.243	0.170	0.217

q	rates quoted after a good signal	rates quoted after a bad signal
0.70	[0.314, 0.395]	[0.395, 0.45]
0.80	[0.302, 0.398] $\cup$ [0.425, 0.45]	[0.398, 0.45]
0.90	[0.308, 0.45]	[0.433, 0.45] $\cup$ [0.511, 0.60]
0.97	[0.297, 0.45]	[0.482, 0.60]

- $q = 0.70$, **ranges meet.** Good-signal banks quote on [0.314, 0.395] and bad-signal banks on [0.395, 0.45]; the ranges touch at $\rho = 0.395$. After a bad signal the firm is still good with probability $\alpha^{sb} = 0.44$, so $\varphi^{sb} > 0$ and bad-signal banks pool: nobody quotes in the bad-only region.
- $q = 0.80$, **good-signal banks pause.** Good-signal banks quote up to 0.398, stop, and place their last 3.3% of quotes on [0.425, 0.45]. On the pause [0.398, 0.425] only bad-signal banks quote; the density panel shows the gap in the blue curve.

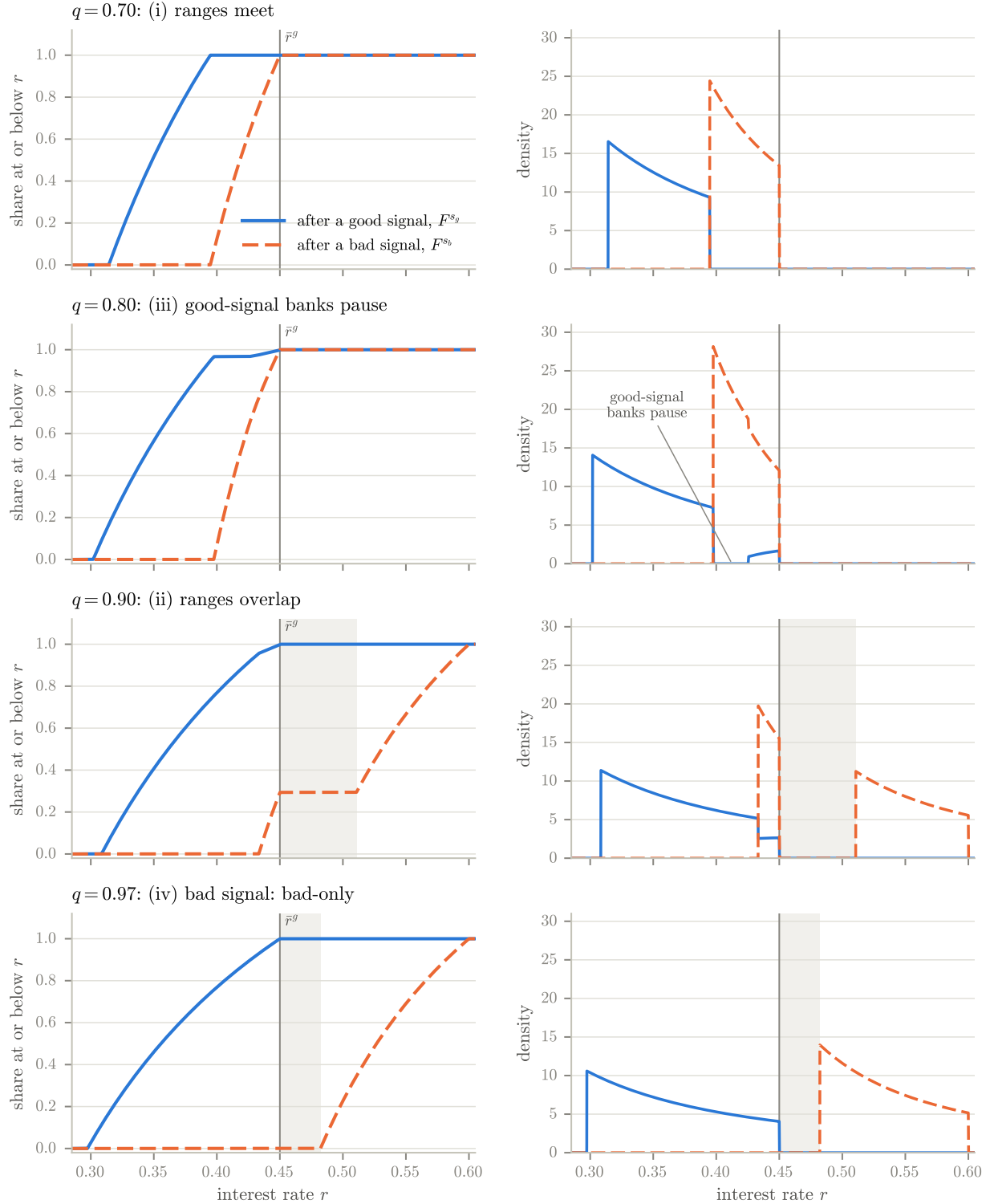


Figure 1: Equilibrium rate distributions at four values of q , one row each. Left: share of quotes at or below each rate. Right: density of quotes. Solid blue: after a good signal (F^{sg}). Dashed orange: after a bad signal (F^{sb}). Vertical line: $\bar{r}^g$, the top of the pooling region. Grey band: rates in the bad-only region that nobody quotes. Economy of Section 5.

- $q = 0.90$, **ranges overlap**. Good-signal banks quote up to $\bar{r}^g$. Bad-signal banks are hybrid: they share $[0.433, 0.45]$ with good-signal banks and place 71% of their quotes in the bad-only region, on $[0.511, 0.60]$. Nobody quotes on $(0.45, 0.511)$.
- $q = 0.97$, **bad-only**. Good-signal banks quote alone in the pooling region, on $[0.297, 0.45]$. Bad-signal banks lend to bad firms only, on $[0.482, 0.60]$.

5.2 Checking the equilibrium at every rate

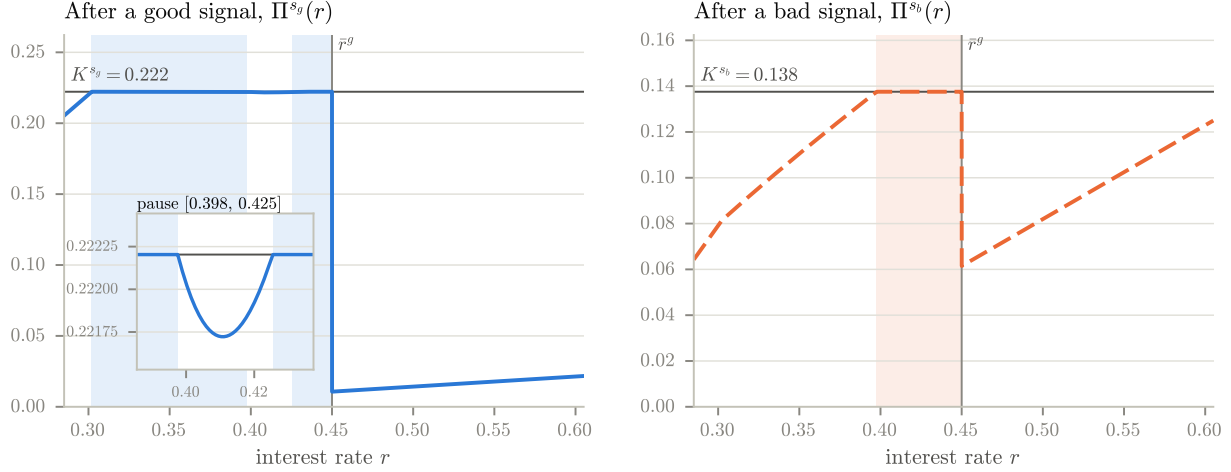


Figure 2: Profit after each signal at every rate, against the equilibrium distributions at $q = 0.80$ (the pause). Shaded: rates quoted after that signal. Horizontal line: equilibrium profit K^s . Vertical line: $\bar{r}^g$. Inset: the pause. Economy of Section 5.

Profit equals K^s wherever the signal quotes and is lower everywhere else.

- **Good signal.** Below the lowest quote (0.302) profit rises with the rate. During the pause it dips below K^{s_g} by 0.0005 (0.2%): too little to see at full scale, but enough that no good-signal bank quotes there. In the bad-only region good firms reject, and profit falls to what bad firms alone pay.
- **Bad signal.** Profit rises until $\rho = 0.398$, stays at K^{s_b} up to $\bar{r}^g$, drops at $\bar{r}^g$ as the good firms in the pool reject, and climbs back only to 0.123 at $\bar{r}^b = 0.60$. That is why nobody quotes in the bad-only region at this q .

5.3 The equilibrium as q rises

- $q < 0.78$: **ranges meet**. Good-signal banks quote the low rates and bad-signal banks the rates up to $\bar{r}^g$. The meeting rate rises from 0.373 to 0.399.
- $0.78 \leq q < 0.83$: **good-signal banks pause**. The pause opens just below $\bar{r}^g$ ($[0.399, 0.449]$ at $q = 0.78$) and shrinks as q rises ($[0.397, 0.405]$ at $q = 0.82$) until it closes.
- $0.83 \leq q < 0.93$: **ranges overlap**. Good-signal banks quote up to $\bar{r}^g$. From $q = 0.834$, $\varphi^{s_b} < 0$: bad-signal banks turn from full pooling to hybrid, m rises from 0 to 1, and their range in the pooling region shrinks toward $\bar{r}^g$.

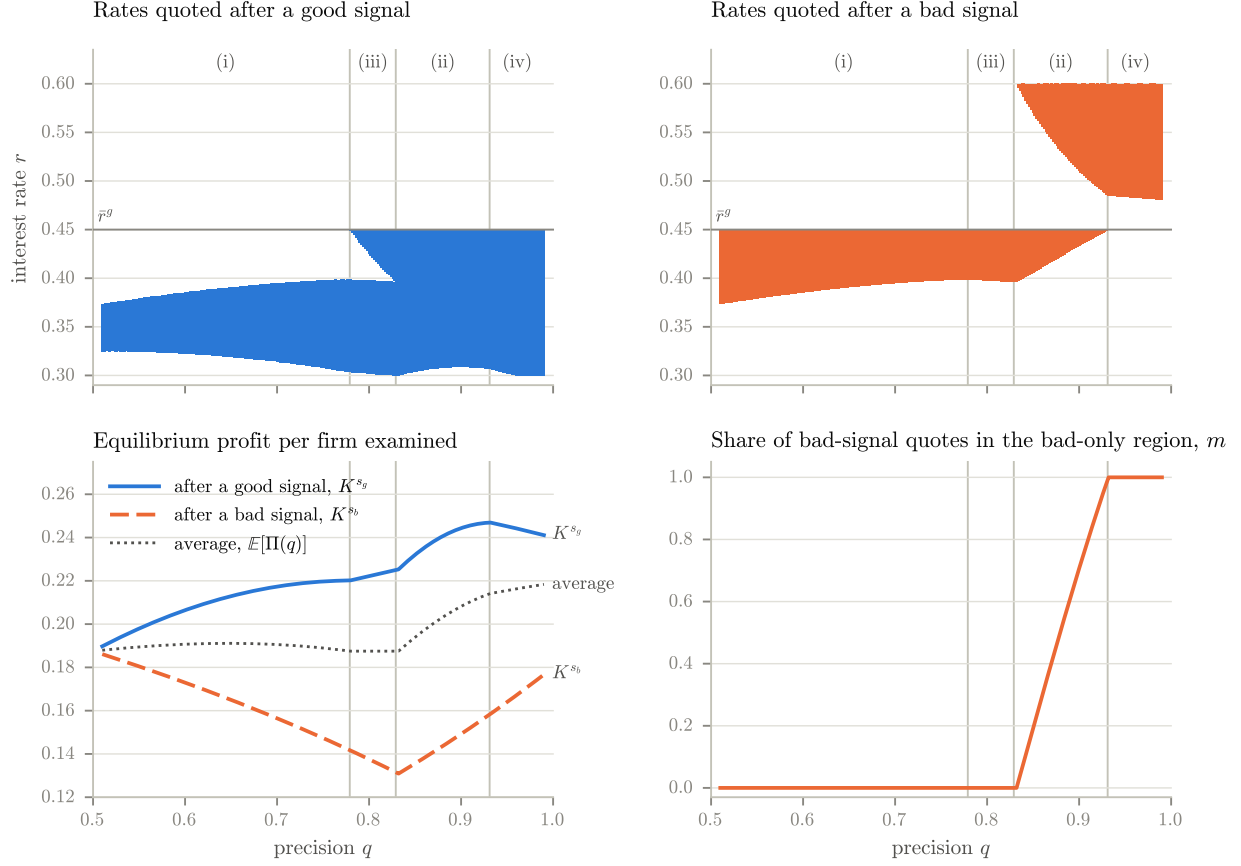


Figure 3: The equilibrium as q rises from 0.51 to 0.99, other parameters fixed. Top: rates quoted after a good signal (left) and after a bad signal (right), with the configuration above each range of q . Bottom left: equilibrium profit after each signal and expected profit. Bottom right: share of bad-signal quotes in the bad-only region. Vertical lines: changes of configuration, at $q = 0.78, 0.83$ and 0.93 . Horizontal line: $\bar{r}^g$. Economy of Section 5.

- $q \geq 0.93$: **bad-only**. Bad-signal banks quote on roughly $[0.48, 0.60]$.

Profits:

- K^{sg} rises from 0.190 at $q = 0.51$ to 0.247 at $q = 0.93$, then falls to 0.241.
 - K^{sb} falls from 0.186 to 0.131 at $q = 0.83$, while bad-signal banks still pool with fewer and fewer good firms. It then rises to 0.177 as they move to the bad-only region.
 - Expected profit per firm examined stays between 0.187 and 0.191 up to $q = 0.83$, then rises to 0.218 at $q = 0.99$.
-

6. Information acquisition: profitable deviations

The draft asks whether a market's precision q^* is an equilibrium of the information-acquisition game: can a bank gain by choosing its own precision q_1 while its rival keeps q^* and the equilibrium rates $F^s(\cdot; q^*)$? With private signals the question has an exact answer.

6.1 The deviating bank's problem

What stays fixed. The rival keeps precision q^* and the rates of the equilibrium at q^* . The two signals are still independent given the firm's type. Holding the rival's rates fixed is the right test when the rival cannot see q_1 before it sets its rates.

What does not change. Given the firm's type, the rival's signal depends only on q^* . So Y_g, Y_b and the values Φ_g, Φ_b are those of the equilibrium at q^* , with $\Phi_g = 0$ in the bad-only region.

What changes. Only the deviator's own shares and beliefs, $\sigma_g(q_1)$, $\alpha^{sg}(q_1)$ and $\alpha^{sb}(q_1)$, from Section 1 with q_1 in place of q . After a signal that leaves belief α , the deviator quotes its best rate and earns

$$W(\alpha) = \max_r [\alpha \Phi_g(r) + (1 - \alpha) \Phi_b(r)].$$

Its expected profit per firm examined is

$$\Pi^{dev}(q_1) = \sigma_g(q_1) W(\alpha^{sg}(q_1)) + \sigma_b(q_1) W(\alpha^{sb}(q_1)), \quad (10)$$

with $\Pi^{dev}(q^*) = \mathbb{E}[\Pi(q^*)]$. The case $q_1 = \frac{1}{2}$ is a bank that does not consult a signal. The draft's participation decision, whether to pay C_s for the signal, compares $\Pi^{dev}(q^*) - C_s$ with $\Pi^{dev}(\frac{1}{2})$.

When the two precisions differ, a firm on which the banks disagree is no longer good with probability μ_g : the signal of the more precise bank counts more. At $q^* = 0.9$, a deviator that drew g while the rival drew b faces a good firm with probability 0.95 if $q_1 = 0.99$ and 0.45 if $q_1 = 0.80$. A bank wins shoppers mostly when the rival disagrees, so a sharper signal improves the firms it wins.

6.2 How deviation profit depends on precision

Let D^+ and D^- denote the slopes of Π^{dev} just above and just below q^* .

Proposition 8. Π^{dev} is convex and non-decreasing in q_1 , and

$$D^\pm = \mu_g[\Phi_g(r_g^\pm) - \Phi_g(r_b^\pm)] - \mu_b[\Phi_b(r_g^\pm) - \Phi_b(r_b^\pm)]. \quad (11)$$

A little more precision makes the good signal more confident and the bad signal less. By (4), after a good signal the deviator moves to the rate with the highest premium in the good-signal range, $r_g^+ = \underline{r}$, and after a bad signal to the rate with the lowest premium in the bad-signal range, r_b^+ . A little less precision does the opposite: r_g^- has the lowest premium in the good-signal range and r_b^- the highest in the bad-signal range. Two closed forms follow:

- If $m > 0$, every rate in the bad-only range has the lowest premium, with $\Phi_g = 0$ and $\Phi_b = (1 - \pi)(\bar{r}^b - c_b)$:

$$D^+ = \mu_g(\underline{r} - c_g) - \mu_b[(\underline{r} - c_b) - (1 - \pi)(\bar{r}^b - c_b)].$$

- In configuration (iv), $r_g^- = \bar{r}^g$ and r_b^- is any rate in the bad-only range:

$$D^- = \mu_g(1 - \pi q^*)(\bar{r}^g - c_g) - \mu_b[(1 - \pi + \pi q^*)(\bar{r}^g - c_b) - (1 - \pi)(\bar{r}^b - c_b)].$$

If $m = 0$, the lowest premium in the bad-signal range need not be at the top of the market. In configuration (iii) it lies inside the pause.

Proposition 9 (less precision). $\Pi^{dev}(q_1) = \Pi^{dev}(q^*)$ for every $q_1 \leq q^*$ if and only if some rate $r^\sharp$ is quoted after both signals in equilibrium, which is the case exactly when $m < 1$, in configurations (i) to (iii). The deviator then quotes $r^\sharp$ after both signals and earns $\mu_g\Phi_g(r^\sharp) + \mu_b\Phi_b(r^\sharp)$, which by (1) does not depend on q_1 . In configuration (iv), $\Pi^{dev}(q_1) < \Pi^{dev}(q^*)$ for every $q_1 < q^*$, although Π^{dev} can be flat over a range of low q_1 .

At a rate quoted after both signals, profit depends only on how many good firms the bank lends to, which is μ_g whatever its precision. Competition has already equalized the two signals' profits there, so the information is worth nothing. For the draft's participation decision: whenever the two signals share a rate, a bank that stops paying for the signal earns what a bank that pays earns. No bank then pays any $C_s > 0$, and a market in which both banks consult the signal unravels.

6.3 Which markets sustain their precision

Let $C(q)$ be the cost of precision q , strictly increasing. The market's precision q^* survives if no bank gains by deviating once the cost is counted:

$$\Pi^{dev}(q_1) - C(q_1) \leq \Pi^{dev}(q^*) - C(q^*) \quad \text{for every } q_1.$$

Proposition 10. For $q^* > \frac{1}{2}$, a strictly increasing convex cost that sustains q^* exists if and only if $\Pi^{dev}(q_1) < \Pi^{dev}(q^*)$ for every $q_1 < q^*$.

The construction is a cost that rises by S_{down} per unit of precision below q^* and by S_{up} above it, with

$$S_{down} = \frac{\Pi^{dev}(q^*) - \Pi^{dev}(\frac{1}{2})}{q^* - \frac{1}{2}}, \quad S_{up} = \frac{\Pi^{dev}(1) - \Pi^{dev}(q^*)}{1 - q^*}.$$

Because Π^{dev} is convex, S_{down} is the smallest loss per unit of precision dropped and S_{up} the largest gain per unit added, so $S_{up} \geq S_{down}$.

Proposition 11. Only two kinds of market sustain their precision against some strictly increasing convex cost: $q^* = \frac{1}{2}$, and configuration (iv), that is $\varphi^{sb} \leq -\frac{\pi}{1-\pi} B^{sb}(\bar{r}^g)$. Markets in which the ranges meet, overlap, or are separated by a pause are ruled out: a bank lowers its precision, quotes the shared rate, loses nothing, and saves cost.

In configuration (iv) no smooth cost works either: Π^{dev} has a corner at q^* with $D^- < D^+$, so a differentiable cost would need a marginal cost at q^* no larger than D^- and no smaller than D^+ . The cost must rise slowly up to q^* and steeply beyond it, as with a screening technology that cannot go past q^* .

6.4 The two ends

Proposition 12 (starting from no information). At $q^* = \frac{1}{2}$, with $\mu_b = 1 - \mu_g$,

$$D^+(\frac{1}{2}) = 2\mu_g\mu_b [W'_+(\mu_g) - W'_-(\mu_g)] = \begin{cases} 2\pi\mu_g\mu_b(c_b - c_g) & \text{full pooling,} \\ 2\mu_g\mu_b[(c_b - c_g) + (1 - \pi)(\bar{r}^b - c_b)] & \text{hybrid,} \\ 0 & \text{bad-only,} \end{cases}$$

where the regime is the draft's at $q = \frac{1}{2}$. The market with no information survives if and only if $C'(\frac{1}{2}) \geq D^+(\frac{1}{2})$. Under full pooling the gain from the first unit of precision is the product of three things: the share of shoppers (without them rates are not dispersed), $\mu_g\mu_b$ (there is nothing to learn about a pool of one type), and $c_b - c_g$ (what is at stake per firm). Under bad-only lending no good firm borrows, so there is nothing to sort and the first unit of precision is worth nothing.

Perfect signal. At $q^* = 1$, if $(1 - \pi)(\bar{r}^b - \bar{r}^g) \geq \pi(\bar{r}^g - c_b)$, bad-signal banks lend in the bad-only region only (Section 4) and

$$D^-(1) = \mu_g(1 - \pi)(\bar{r}^g - c_g) - \mu_b[(\bar{r}^g - c_b) - (1 - \pi)(\bar{r}^b - c_b)].$$

A perfect signal survives if $C'(1^-) \leq D^-(1)$ (and the whole saving from any lower precision stays below the profit lost). If instead $(1 - \pi)(\bar{r}^b - \bar{r}^g) < \pi(\bar{r}^g - c_b)$, the good-type and bad-type ranges share rates, Π^{dev} is flat on $[\frac{1}{2}, 1]$, and no increasing cost sustains a perfect signal.

In the economy of Section 5, $D^+(\frac{1}{2}) = 0.0455$ (full pooling) and $D^-(1) = 0.167$; a bank that drops from $q^* = 1$ to no information loses 4.8% of profit.

6.5 Simulation: deviations from four equilibria

The rival sits at each of the four equilibria of Section 5.1. Gains and losses are in % of $\Pi^{dev}(q^*)$, before the cost of precision. The script writes `figures/dev_curves.*` and `deviation_curves.csv`.

q^*	configuration	$\Pi^{dev}(q^*)$	lower to $q_1 = 0.5$	raise to $q_1 = 0.999$	D^-	D^+
0.70	(i) ranges meet	0.191	0%	+23.1%	0	0.042
0.80	(iii) pause	0.188	0%	+20.8%	0	0.039
0.90	(ii) ranges overlap	0.208	0%	+10.8%	0	0.228
0.97	(iv) bad-only	0.217	-2.8%	+3.0%	0.170	0.225

- **Lowering precision.** In configurations (i) to (iii) profit does not change. The deviator quotes a shared rate after both signals: 0.395 at $q^* = 0.70$, 0.398 at $q^* = 0.80$, and 0.433 at $q^* = 0.90$. At $q^* = 0.97$ no rate is shared. For q_1 below 0.935 the deviator quotes $\bar{r}^g$ after both signals and loses 2.8%; between 0.935 and 0.97 it quotes $\bar{r}^g$ after a good signal and 0.482 after a bad one, and the loss shrinks to zero.
- **Raising precision.** After a good signal the deviator quotes the lowest rate in the market: 0.314, 0.302, 0.308 and 0.297. After a bad signal, at $q^* = 0.70$ it quotes $\bar{r}^g$, and at $q^* = 0.80$ it quotes 0.411, inside the pause (Proposition 8). In both markets it moves to $\bar{r}^b = 0.60$ from $q_1 = 0.83$ on, lending to bad firms only, which is the kink in the gain. At $q^* = 0.90$ and 0.97 it quotes in the bad-only range (0.511 and 0.482). The slopes in the table match (11).

6.6 Simulation: deviations at every market precision

- The gain from raising precision to 0.999 falls from 28.5% of profit at $q^* = 0.51$ to 7.3% at $q^* = 0.93$ and 0.9% at $q^* = 0.99$.
- The loss from lowering precision to 0.5 is zero up to $q^* = 0.930$ (configurations (i) to (iii)). It then rises to 4.1% at $q^* = 0.99$, and exceeds the gain from raising from $q^* = 0.972$ on.
- S_{up} stays between 0.11 and 0.23 at every q^* . S_{down} is zero up to $q^* = 0.930$ and stays below 0.02.
- In configuration (iv), $D^+ - D^-$ is at least 0.05 at every $q^* \geq 0.932$: a cost that sustains these markets must have a marginal cost that jumps at q^* by at least that much.

7. Checks

`check_formulas.py` (same folder) tests the propositions on random economies:

- **Equilibrium.** 482 of 500 economies satisfy (A). In each, profit equals K^s on each support and no rate earns more, to 3×10^{-15} . Each of the other 18 has a rate that earns more than K^s . All four configurations occur (157, 122, 27 and 194 economies).

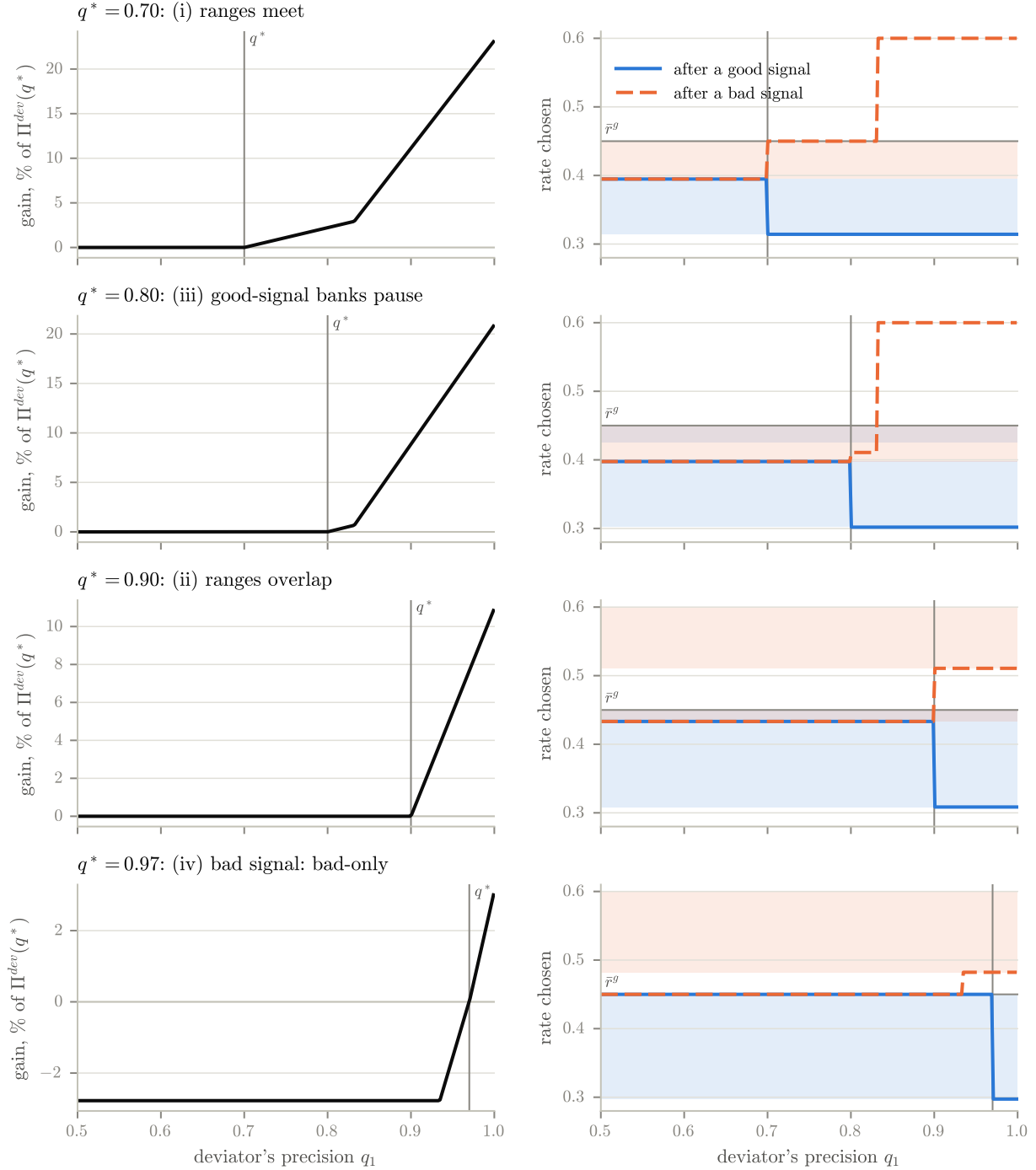


Figure 4: One bank deviates to precision q_1 while the rival stays at the equilibrium with precision q^* (vertical line), for the four equilibria of Section 5.1. Left: change in deviation profit, $\Pi^{dev}(q_1) - \Pi^{dev}(q^*)$, in % of $\Pi^{dev}(q^*)$, before the cost of precision. Right: rate the deviator quotes after a good signal (solid blue) and after a bad signal (dashed orange); when several rates tie, the lowest. Light bands: rates quoted in equilibrium after a good signal (blue) and after a bad signal (orange); where the bands overlap, both signals quote. Horizontal line: $\bar{r}^g$. Economy of Section 5.

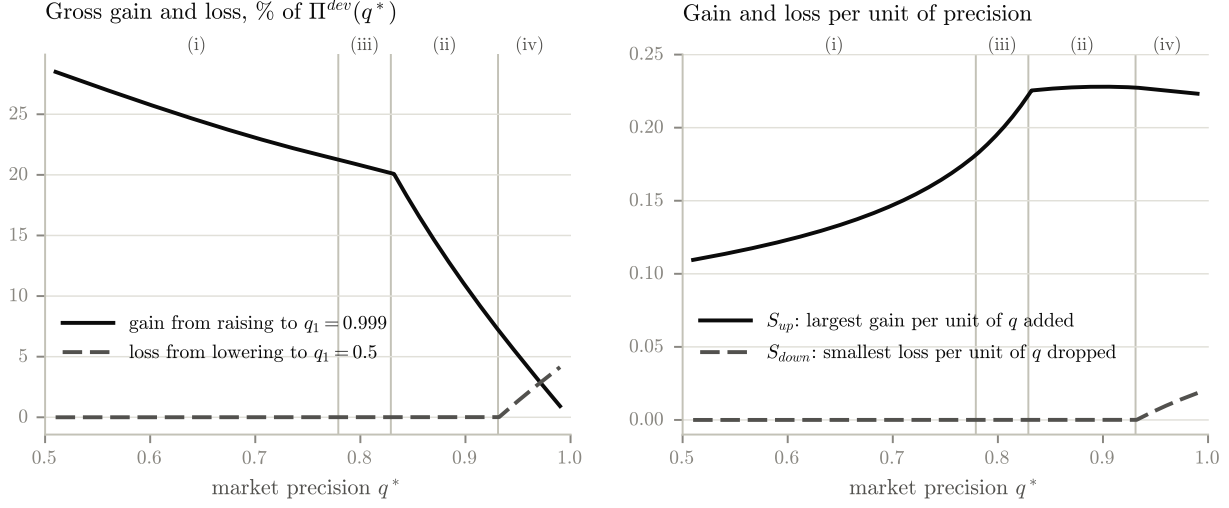


Figure 5: Gains and losses from a one-bank deviation, at every equilibrium precision q^* of the Section 5 economy. Left: gain from raising precision to $q_1 = 0.999$ and loss from lowering it to $q_1 = 0.5$, in % of $\Pi^{dev}(q^*)$, before the cost of precision. Right: S_{up} and S_{down} , with $q_1 = 0.999$ in place of 1. Vertical lines and labels: configurations, as in Figure 3. Economy of Section 5.

- **Rates.** The quadratic for ρ matches the construction to 10^{-12} . The limit of m as $q \rightarrow 1$ matches Section 4.
- **Near no information.** At $q = 0.52$, condition (A) holds in 666 of 666 economies that pool fully at $q = \frac{1}{2}$, in 194 of 236 hybrid ones, and in 0 of 98 bad-only ones.
- **Deviations** (143 economies satisfying (A)). The slopes (11) match finite differences to 3×10^{-7} . Proposition 9 holds in all 143; in 45 economies in configuration (iv), Π^{dev} is flat over a range of low q_1 . Proposition 12 matches in 300 economies (180 full pooling, 90 hybrid, 30 bad-only) to 10^{-9} . The perfect-signal slope matches in 131 economies; in the other 69, Π^{dev} is flat down to $q_1 = \frac{1}{2}$.

8. Differences from note.pdf

- **Setting and notation** are the draft's: two banks, the pooling and bad-only regions, φ^s , $c^{avg,s}$, and the regimes full pooling, hybrid and bad-only.
- **Condition (A).** The note's Proposition 4 misses the case in which both signals lend in the bad-only region only. Section 3 assumes (A) and states when it fails.
- **Limit** $q \rightarrow 1$. The note states $m = 1$. In general $m \rightarrow \min\{1, (1 - \pi)(\bar{r}^b - \bar{r}^g)/(\pi(\bar{r}^g - c_b))\}$.
- **Rate** ρ . The note's equation for configuration (ii) holds for $m = 0$; Section 3.8 gives it for every m . The test $\rho_g < \rho$ for configuration (i) and the closed forms (8) restate the note's construction.

- **Proposition 8.** The note sends the more precise bank to the highest rate in the market after a bad signal. That is right when $m > 0$. When $m = 0$ the right rate is the one with the lowest premium in the bad-signal range, which in configuration (iii) lies inside the pause.
 - **Proposition 9.** In configuration (iv) the note states that Π^{dev} rises strictly below q^* . It is lower than $\Pi^{dev}(q^*)$ at every $q_1 < q^*$ but can be flat at low q_1 . Propositions 10 and 11 are unaffected.
 - **Proposition 12.** The note covers full pooling and bad-only lending; the hybrid regime is added. The note's slope at $q^* = 1$ holds only when $(1 - \pi)(\bar{r}^b - \bar{r}^g) \geq \pi(\bar{r}^g - c_b)$.
-

Appendix A. Proof sketches

Lemma 1. Profit is the expectation over the firm's type of the chance of keeping the firm times the margin. Given the type, the rival's signal is g with probability q for a good firm and $1 - q$ for a bad firm, whatever the bank drew; averaging the rival's distribution over its signal gives Y_θ . A captive firm accepts and a shopper accepts when not undercut, so the chance of keeping a type- θ firm is $1 - \pi + \pi Y_\theta$ and the margin is $r - c_\theta$. The bank's own signal enters only through the weights α^s . In the bad-only region good firms reject, which deletes the first term.

Lemma 2. Equation (4) is Lemma 1 differenced. Where F^{sg} has mass, $\Pi^{sg} = K^{sg}$ and $\Pi^{sb} \leq K^{sb}$, so $(\alpha^{sg} - \alpha^{sb})\Delta \geq K^{sg} - K^{sb}$. The same argument gives the second claim, and the two together give the third. Where both have mass, the two conditions $\alpha^s \Phi_g + (1 - \alpha^s)\Phi_b = K^s$ are linear in (Φ_g, Φ_b) with determinant $\alpha^{sg} - \alpha^{sb} \neq 0$, so their solution is unique and does not depend on r .

Lemma 3. Substitute Y_g and Y_b into $\Delta = \Phi_g - \Phi_b$ and collect the terms in F^{sg} and F^{sb} . $G_1 > 0$ because $q > 1 - q$ and $r - c_g > r - c_b$. G_2 is linear with slope $-(2q - 1)$ and is zero at $r^\dagger$.

Lemma 4. Suppose nobody quotes on (a, b) in the pooling region while a is in some support. Moving those quotes from a to just below b leaves Y_g and Y_b unchanged. With Y_g, Y_b fixed, the derivative of Π^s in r is $\alpha^s[1 - \pi + \pi Y_g] + (1 - \alpha^s)[1 - \pi + \pi Y_b] > 0$, so the move raises profit above K^s , which contradicts equilibrium.

Proposition 1. Let R_g be the top of the good-signal range, where $F^{sg} = 1$. By Lemma 3, $\Delta(R_g) = (c_b - c_g) - \pi[G_1 + F^{sb}G_2]$, and $G_1 + F^{sb}G_2 \geq \min\{G_1, G_1 + G_2\} > 0$ because $G_1 + G_2 = c_b - c_g$. So $\Delta(R_g) < c_b - c_g$, and Lemma 2 gives $\Delta^* \leq \Delta(R_g) < c_b - c_g$. At the lowest rate both distributions are zero, so $\Delta = c_b - c_g > \Delta^*$: by Lemma 2 bad-signal banks cannot quote there.

Proposition 2. Suppose the good-signal top R_g exceeds the bad-signal top R_b . At R_g nobody quotes higher, so $\Delta(R_g) = (1 - \pi)(c_b - c_g)$. At R_b , $F^{sb} = 1$ and $F^{sg} = 1 - u$ with $u > 0$, so $\Delta(R_b) = (1 - \pi)(c_b - c_g) + \pi u G_1(R_b) > \Delta(R_g)$. Both signals quote near R_b , so $\Delta(R_b) = \Delta^*$ and $\Delta(R_g) < \Delta^*$. But only good-signal banks quote on $(R_b, R_g]$, where Lemma 2 requires $\Delta \geq \Delta^*$: a contradiction.

Proposition 3. Let R be the highest rate in the market; by Proposition 2 it is quoted after a bad signal, and nobody quotes above it. If $R < \bar{r}^g$, quoting slightly above R earns $(1 - \pi)(R + \epsilon - c^{avg, sb}) > K^{sb}$; the same holds inside the bad-only region. So $R \in \{\bar{r}^g, \bar{r}^b\}$, and Lemma 1 with $Y_g = Y_b = 0$ gives (5). $R = \bar{r}^g$ requires that quoting $\bar{r}^b$ earn no more, $(1 - \alpha^{sb})(1 - \pi)(\bar{r}^b - c_b) \leq (1 - \pi)(\bar{r}^g - c^{avg, sb})$,

which is $\varphi^{s_b} \geq 0$. Expanding $c^{avg,s}$ gives the second form of (6).

Proposition 4. In the bad-only region the value of a bad firm is at most $(1 - \pi)(\bar{r}^b - c_b)$: that is its value on the bad-signal range there, and it is lower on the band nobody quotes. A good-signal bank therefore earns at most $(1 - \alpha^{s_g})(1 - \pi)(\bar{r}^b - c_b)$ there, which (A) rules out. With every good-signal quote below, $Y_b = q(1 - F^{s_b})$, and $(1 - \alpha^{s_b})[1 - \pi + \pi q(1 - F^{s_b})](r - c_b) = K^{s_b}$ with K^{s_b} from (5) gives the distribution; $1 - F^{s_b}(\underline{r}^b) = m$ gives $\underline{r}^b$. At $\bar{r}^g$, $F^{s_g} = 1$ and $F^{s_b} = 1 - m$, so by (3), using $A^s + B^s = r - c^{avg,s}$, $\Pi^{s_b}(\bar{r}^g) = (1 - \pi)(\bar{r}^g - c^{avg,s_b}) + \pi m B^{s_b}(\bar{r}^g)$. Setting this equal to $K^{s_b} = (1 - \pi)(\bar{r}^g - c^{avg,s_b}) - (1 - \pi)\varphi^{s_b}$ gives (7), bounded by $m \in [0, 1]$. On $(\bar{r}^g, \underline{r}^b)$ a bad-signal bank earns $(1 - \alpha^{s_b})[1 - \pi + \pi q m](r - c_b)$, which reaches K^{s_b} only at $\underline{r}^b$.

Proposition 5. By Lemma 2, Φ_g and Φ_b are constant on a range both signals quote. In each configuration of Proposition 7 that range ends at $\bar{r}^g$, where $F^{s_g} = 1$ and $F^{s_b} = 1 - m$ give $Y_g = (1 - q)m$ and $Y_b = qm$, so the constants are $\Phi_g^\top$ and $\Phi_b^\top$. Inverting (2) gives $1 - Y_\theta = [(r - c_\theta) - \Phi_\theta^\top]/[\pi(r - c_\theta)]$, and inverting the two equations for Y_g, Y_b gives (8). Where only good-signal banks quote, $F^{s_b} = 0$ and (3) is linear in F^{s_g} , which gives (9); $F^{s_g} = 0$ gives $\underline{r}$.

Proposition 6. Differentiating (8), $\pi(2q - 1)F^{s_g'} = q\Phi_g^\top/(r - c_g)^2 - (1 - q)\Phi_b^\top/(r - c_b)^2$ and $\pi(2q - 1)F^{s_b'} = q\Phi_b^\top/(r - c_b)^2 - (1 - q)\Phi_g^\top/(r - c_g)^2$. So $F^{s_g'} \geq 0$ if and only if $[(r - c_b)/(r - c_g)]^2 \geq \tau^2$, that is $r \geq \rho_-$, because $(r - c_b)/(r - c_g)$ rises with r . $F^{s_b'} \geq 0$ if and only if $(\Phi_g^\top/\Phi_b^\top)[(r - c_b)/(r - c_g)]^2 \leq q/(1 - q)$. The left side rises with r and at $\bar{r}^g$ equals $\frac{1 - \pi + \pi(1 - q)m}{1 - \pi + \pi q m} \cdot \frac{\bar{r}^g - c_b}{\bar{r}^g - c_g} < 1 < \frac{q}{1 - q}$, so this always holds. At $\bar{r}^g$, $F^{s_g} = 1$ by construction.

Proposition 7. By Proposition 1 the market opens with good-signal banks alone, on (9). That phase ends when their quotes run out, at ρ_g , or when bad-signal banks become willing to quote. At the latter rate both indifference conditions hold with $F^{s_b} = 0$, so (9) and (8) agree there: it is ρ . If $\rho_g < \rho$, nobody would quote just above ρ_g , contradicting Lemma 4. Raising K^{s_g} lowers (9) at every rate until good-signal quotes end where bad-signal quotes begin, with $F^{s_g} = 1$ and $F^{s_b} = 0$; then $1 - \pi + \pi Y_g = L$ and $1 - \pi + \pi Y_b = M$, and $\Pi^{s_b}(\rho) = K^{s_b}$, $\Pi^{s_g}(\rho) = K^{s_g}$ give configuration (i). Above ρ , (3) with $F^{s_g} = 1$ is linear in F^{s_b} , and $(r - c^{avg,s_b}) - \pi A^{s_b}(r) = \alpha^{s_b}L(r - c_g) + (1 - \alpha^{s_b})M(r - c_b)$. If $\rho_g \geq \rho$, the two branches meet at ρ with $F^{s_g} < 1$ and $K^{s_g} = K^\circ$. If $\rho \geq \rho_-$, F^{s_g} rises on (8) from ρ on: configuration (ii). If $\rho < \rho_-$, sharing would need F^{s_g} to fall, so good-signal banks stop at $\bar{F}$ and only bad-signal banks quote, with (3) linear in F^{s_b} . Sharing resumes where (8) returns to $\bar{F}$; by Proposition 6, (8) falls until ρ_- and then rises to one, so it crosses $\bar{F} < 1$ once more, at $r_2 > \rho_-$. Above r_2 , F^{s_g} rises, so there is no second pause: configuration (iii). If $m = 1$, no bad-signal bank quotes in the pooling region, and (9) with $K^{s_g} = K^\circ$ reaches one at $\bar{r}^g$: configuration (iv).

Proposition 8. Since $\sigma_g \alpha^{s_g} = q_1 \mu_g$, $\sigma_g(1 - \alpha^{s_g}) = (1 - q_1)\mu_b$, $\sigma_b \alpha^{s_b} = (1 - q_1)\mu_g$ and $\sigma_b(1 - \alpha^{s_b}) = q_1 \mu_b$, quoting r_g after a good signal and r_b after a bad one earns $q_1 \mu_g \Phi_g(r_g) + (1 - q_1)\mu_b \Phi_b(r_g) + (1 - q_1)\mu_g \Phi_g(r_b) + q_1 \mu_b \Phi_b(r_b)$, which is linear in q_1 . Π^{dev} is the largest of these lines, so it is convex, and its one-sided slopes are the slopes of the lines through the best rates, (11). On the support of F^s , $\alpha \Phi_g + (1 - \alpha)\Phi_b = K^s + (\alpha - \alpha^s)\Delta$, so for a belief just above α^s the best rate in the support maximizes Δ , and just below it minimizes Δ . Just above q^* the good-signal belief rises and the bad-signal belief falls, which gives r_g^+ and r_b^+ ; just below, the reverse. On the good-signal range Δ is largest at $\underline{r}$, where it equals $c_b - c_g$. For monotonicity, write $X = \mu_g[\Phi_g(r_g) - \Phi_g(r_b)]$ and $Z = \mu_b[\Phi_b(r_g) - \Phi_b(r_b)]$. Optimality of r_g after a good signal and of r_b after a bad one give $q_1 X + (1 - q_1)Z \geq 0$ and $(1 - q_1)X + q_1 Z \leq 0$. Subtracting, $(2q_1 - 1)(X - Z) \geq 0$, so the slope $X - Z$ is non-negative.

Proposition 9. W is the largest of functions linear in α , so it is convex. If $r^\sharp$ is best at both

$\alpha^{sb}(q^*)$ and $\alpha^{sg}(q^*)$, it is best at every belief in between. Less precision moves both beliefs inside that interval, so the deviator quotes $r^\sharp$ after both signals and, by (1), earns $\mu_g \Phi_g(r^\sharp) + \mu_b \Phi_b(r^\sharp)$. If no rate is best at both ends, W is not linear on the interval, so it lies strictly below its chord inside the interval. $\Pi^{dev}(q_1)$ is an average of W at two interior beliefs with mean μ_g , which is below the chord at μ_g , that is below $\Pi^{dev}(q^*)$. If $m < 1$, the meeting rate (configuration (i)) or any rate on a shared range (configurations (ii) and (iii)) is quoted after both signals. If $m = 1$, the good-signal range ends at $\bar{r}^g$ and the bad-signal range starts at $\underline{r}^b > \bar{r}^g$, so no rate is shared.

Proposition 10. Necessity: if q^* is best and C is strictly increasing, then for $q_1 < q^*$, $\Pi^{dev}(q^*) - \Pi^{dev}(q_1) \geq C(q^*) - C(q_1) > 0$. Sufficiency: take C piecewise linear through q^* with slope S_{down} below and S_{up} above. Both slopes are positive, and convexity of Π^{dev} gives $S_{up} \geq S_{down}$, so C is strictly increasing and convex. Convexity also makes the chord slope $[\Pi^{dev}(q^*) - \Pi^{dev}(q_1)]/(q^* - q_1)$ rise with q_1 , so it is at least S_{down} , and $C(q^*) - C(q_1) \leq \Pi^{dev}(q^*) - \Pi^{dev}(q_1)$. The argument above q^* is the same.

Proposition 11. At $q^* = \frac{1}{2}$ there is no lower precision, and a steep enough cost deters any increase. For $q^* > \frac{1}{2}$, Proposition 9 gives a flat Π^{dev} below q^* when $m < 1$, which violates Proposition 10, and $\Pi^{dev}(q_1) < \Pi^{dev}(q^*)$ below q^* when $m = 1$, which satisfies it. By (7), $m = 1$ is $\varphi^{sb} \leq -\frac{\pi}{1-\pi} B^{sb}(\bar{r}^g)$.

Proposition 12. At $q_1 = \frac{1}{2}$, $\sigma_g = \sigma_b = \frac{1}{2}$, both beliefs equal μ_g , and $d\alpha^{sg}/dq_1 = -d\alpha^{sb}/dq_1 = 4\mu_g\mu_b$. Differentiating (10), the terms in $W(\mu_g)$ cancel and $D^+(\frac{1}{2}) = 2\mu_g\mu_b[W'_+(\mu_g) - W'_-(\mu_g)]$. By the argument in Proposition 8, $W'_+(\mu_g)$ and $W'_-(\mu_g)$ are the largest and smallest premium over the rates quoted at $q = \frac{1}{2}$. With one distribution F , $\Delta = (1 - \pi F)(c_b - c_g)$ in the pooling region and $\Delta = -\Phi_b$ in the bad-only region. The largest premium is $c_b - c_g$, at the lowest rate. Under full pooling the smallest is $(1 - \pi)(c_b - c_g)$, at $\bar{r}^g$. Under hybrid lending it is $-(1 - \pi)(\bar{r}^b - c_b)$, anywhere in the bad-only range. Under bad-only lending every quoted rate has premium $-(1 - \pi)(\bar{r}^b - c_b)$, so the difference is zero.

Perfect signal. At $q^* = 1$ in configuration (iv), apply the closed form for D^- in Proposition 8 with $q^* = 1$.